

Exam Behaviour Analysis – Insights, Findings & Technical Documentation

Executive Summary

This analysis was conducted on event-level examination logs containing over 8 million event records generated by approximately 88,000 candidates. The objective was to analyze candidate interaction patterns, review behaviour, language distribution, response-state trends, section-wise activity, session duration, and temporal event patterns while distinguishing candidate-driven actions from system-generated events.

The analysis was performed using PostgreSQL for data processing and Power BI for visualization and reporting.

Dataset Overview

Metric	Value
Total Candidates	88,000+
Total Event Records	8.2M+
Data Source	candidate_log
Examination Window	09:00 AM – 07:15 PM

Data Validation & Quality Checks

Before analysis, several validation checks were performed to ensure data quality and consistency.

Validation Checks Performed

1. Total Record Verification

Verified successful restoration and completeness of the dataset.

2. Distinct Candidate Validation

Validated the number of unique candidates appearing in the examination logs.

3. Activity Distribution Validation

Reviewed event distribution across all activity types.

4. Question Section Validation

Validated section identifiers and interaction counts.

5. Question Language Validation

Validated English and Hindi language records.

6. Question Response Validation

Verified available response-state values (A, B, C, D).

7. Missing Value Inspection

Checked key fields for missing or null values.

8. Composite Key Validation

To ensure record uniqueness, a composite key consisting of:

(log_id + candidate_id)

was validated.

SQL Used:

```
SELECT log_id, candidate_id, COUNT() FROM candidate_log GROUP BY log_id, candidate_id HAVING COUNT() > 1;
```

Result: No duplicate composite-key combinations were identified.

Key Findings

1. Candidate Participation

More than 88,000 candidates participated in the examination and generated over 8 million event records.

Insight

The dataset provides a sufficiently large sample size for behavioural and operational analysis.

2. Auto Save Event Analysis

Auto Save events accounted for approximately 90.73% of all recorded events.

Insight

The majority of logged events were system-generated rather than direct candidate actions.

Interpretation

Behavioural conclusions should therefore focus primarily on manual interactions rather than total event counts.

3. Candidate Engagement Segmentation

Candidates were segmented into:

- Light Engagement
- Medium Engagement
- Heavy Engagement

Approximately 96.56% of candidates were classified within the Heavy Engagement category.

Important Limitation

Because Auto Save events constitute most recorded events, engagement categories are heavily influenced by system-generated activity.

Interpretation

These segments should be interpreted as event-volume categories rather than direct measures of candidate engagement.

4. Review Behaviour Analysis

Review-related actions were analyzed separately.

Activity	Events
Mark for Review & Next	536,510
UnMark for Review & Next	227,954

Findings

- Mark for Review actions occurred more than twice as often as UnMark actions.
 - Candidates actively utilized review functionality.
 - Review activity suggests strategic question navigation behaviour.
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5. Section-Wise Interaction Analysis

Interaction volumes were analyzed across examination sections.

Findings

- Section 1 recorded the highest interaction volume.
- Section 4 also generated substantial activity.
- Sections 2 and 3 recorded comparatively lower interaction volumes.

Interpretation

Higher interaction volume indicates a greater number of recorded events and should not automatically be interpreted as higher engagement.

6. Language Distribution Analysis

Language usage was analyzed overall and by section.

Findings

- English interactions exceeded Hindi interactions across all sections.
- Hindi contributed a substantial share of interactions in every section.
- Language distribution varied across examination sections.

Interpretation

Section-level language analysis provides more meaningful insight than overall language percentages alone.

7. Response-State Analysis

Response-state distributions were examined across answer options A, B, C, and D.

Findings

- Option A appeared most frequently in recorded answer states.
- Option D appeared least frequently in recorded answer states.

Important Limitation

The dataset contains no answer key.

Additionally, Auto Save events dominate the dataset.

Interpretation

Response-state counts should not be interpreted as:

- Candidate preference
- Answer popularity
- Answer correctness

The analysis only reflects recorded response states.

8. Hourly Event Trend Analysis

Hourly event volumes were analyzed throughout the examination period.

Findings

- Event volume peaked at approximately 16:00.
- Elevated activity was also observed around 09:00 and 12:00–13:00.
- Event volume declined substantially after 17:00.

Interpretation

Hourly peaks represent overall event-volume trends and include significant Auto Save activity. They should not be interpreted as direct measures of candidate behaviour.

9. Session Duration Analysis

Session duration calculations were incorporated into the analysis.

Findings

Session duration metrics provide additional context regarding candidate participation patterns and complement event-based behavioural analysis.

Dashboard Design Rationale

KPI Cards

Purpose: Provide immediate visibility into key metrics.

Used For:

- Total Candidates
- Total Events
- Auto Save Percentage
- Active Interaction Percentage

Reason: KPI cards allow stakeholders to quickly understand high-level performance indicators.

Clustered Bar Charts

Used For:

- Activity Distribution
- Response-State Distribution
- Review Behaviour Analysis

Reason: Bar charts provide the clearest comparison between discrete categories and make differences easy to interpret.

Stacked Column Chart

Used For:

- Section-wise Language Distribution

Reason: A stacked column chart enables simultaneous comparison of:

- Total interaction volume
- English interactions
- Hindi interactions

across multiple examination sections.

Line Chart

Used For:

- Hourly Event Trend Analysis

Reason: Line charts effectively visualize changes over time and help identify event-volume trends and peak periods.

Slicers

Used For: Interactive filtering and dashboard navigation.

Reason: Improves usability and allows focused exploration of the dataset.

SQL Processing Workflow

Step 1

Restore PostgreSQL dump file.

Step 2

Validate dataset integrity.

Step 3

Generate activity distribution analysis.

Step 4

Generate review behaviour analysis.

Step 5

Create candidate segmentation outputs.

Step 6

Generate section interaction analysis.

Step 7

Generate section-wise language distribution.

Step 8

Generate response-state distributions.

Step 9

Generate hourly event trend analysis.

Step 10

Calculate session duration metrics.

Step 11

Validate composite-key uniqueness.

Power Query Steps

1. Import SQL output files into Power BI.
 2. Validate data types.
 3. Rename columns for consistency.
 4. Remove unnecessary fields.
 5. Apply formatting and cleansing.
 6. Load transformed tables into the Power BI model.
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DAX Measures Used

Total Candidates

Total Candidates = SUM(candidate_segmentation[candidates])

Total Events

Total Events = SUM(activity[total_events])

Auto Save Percentage

Auto Save % = $\text{DIVIDE}(\text{CALCULATE}(\text{SUM}(\text{activity}[\text{total_events}]), \text{activity}[\text{activity}] = \text{"Auto Save"}), \text{SUM}(\text{activity}[\text{total_events}]))$

Active Interaction Percentage

Active Interaction % = $1 - [\text{Auto Save \%}]$

Assumptions & Limitations

1. Auto Save events were treated as system-generated interactions.
 2. No answer key was available for correctness analysis.
 3. Event volume does not necessarily indicate candidate engagement.
 4. Response-state distributions should not be interpreted as preference or popularity.
 5. Hourly event peaks may be influenced by system-generated logging activity.
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Recommendations

Recommendation 1

Prioritize manual interaction events when evaluating candidate behaviour.

Recommendation 2

Continue monitoring review-related actions as indicators of candidate decision-making.

Recommendation 3

Analyze section-level language patterns to understand examination dynamics.

Recommendation 4

Interpret event-volume peaks cautiously due to the dominance of Auto Save events.

Conclusion

The examination generated a substantial volume of event data, with Auto Save activity accounting for most recorded events. Review behaviour, section-wise language distribution, response-state analysis, session duration metrics, and hourly event trends provide meaningful insight into examination activity. The analysis highlights the importance of separating candidate-driven actions from system-generated events to ensure accurate interpretation of behavioural patterns.